Lab Assignment Week 08

CSC 3320 – System-level Programming

Wednesday March 5th, 2026

Introduction

Welcome to the eighth programming lab of CSC 3320! Today, we will be covering the following topics:

1. Functions
2. Arrays
3. Pointers

Lab Policies

- Attendance is mandatory.
- Labs must be completed **individually.**
- TAs are here to help you. Ask them for help!
- Lab assignments are due at midnight on the day of your lab.

Deliverables:

1. The C Code for your program. (.c file).
2. A screenshot of the output in the Terminal.

If you have any questions, please do not hesitate to ask your TA.

Program: Bubble Sort

Write a C program that reads a series of integers from the terminal and implements the bubble sort algorithm as a function. **Your program should prompt the user to enter 10 integers.** Bubble sort is a simple sorting algorithm that works by comparing the adjacent elements in the list and swapping them if the elements are not in the specified order. It is an in-place and stable sorting algorithm that can sort items in data structures such as arrays and linked lists.

You will need to implement this familiar algorithm as a function in your program. The function prototype in your solution should be as follows:

```
void bubble_sort(int array[], int n, int *min, int *max)
```

Your bubble sort should not only sort the array of 10 integers, but also provide the smallest and largest values in the array. The bubble_sort() function must not return any values, but instead sorts the array in place and assigns the *min* and *max* values to their respective pointer arguments. **Hint: The first element in your sorted array will be the min and the last element in your sorted array will be the max.**

Finally, your program should output the sorted list, and the min and max variables to the terminal. See example output below.

Example Output

```
Please Enter Number 1: 22
Please Enter Number 2: 77
Please Enter Number 3: 4
Please Enter Number 4: 28
Please Enter Number 5: 3
Please Enter Number 6: 44
Please Enter Number 7: 93
Please Enter Number 8: -2
Please Enter Number 9: 14
Please Enter Number 10: 15

[-2, 3, 4, 14, 15, 22, 28, 44, 77, 93]
Largest: 93
Smallest: -2
```

Deliverables

For today's lab, you will need to upload the C program code for your bubble sort program and its output in the terminal on iCollege. Please name your C code and screenshot as follows:

- C Files
  - lastname_firstname_filename.c
  - For example: **turing_alan_bubble_sort.c**
- Screenshots
  - lastname_firstname_filename.png
  - For example: **turing_alan_ bubble_sort.png**